

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

Step 1 Database Setup

Query 1: Create the database

```
[mysql> CREATE DATABASE IF NOT EXISTS bookflow_db;  
Query OK, 1 row affected (0.00 sec)  
]
```

Query 2: Select the database

```
[mysql> USE bookflow_db;  
Database changed  
]
```

Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table

```
mysql> CREATE TABLE books (  
->     book_id INT AUTO_INCREMENT PRIMARY KEY,  
->     title VARCHAR(255) NOT NULL,  
->     isbn VARCHAR(13) NOT NULL UNIQUE,  
->     published_year INT,  
[ ->     CONSTRAINT chk_published_year CHECK (published_year < 2027)  
-> );  
Query OK, 0 rows affected (0.00 sec)
```

Query 4: Create the members table

```
mysql> CREATE TABLE members (  
->     member_id INT AUTO_INCREMENT PRIMARY KEY,  
->     full_name VARCHAR(100) NOT NULL,  
->     email VARCHAR(150) NOT NULL UNIQUE  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

Query 5 (verification): Inspect the table structures

```
mysql> DESCRIBE books;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> DESCRIBE members;
```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

```
3 rows in set (0.00 sec)
```

Step 3 Initial Seeding (Valid Data)

Query 6: Insert 3 books

```
mysql> INSERT INTO books (title, isbn, published_year) VALUES  
-> ('The Alchemist', '9780061122415', 1988),  
-> ('Clean Code', '9780132350884', 2008),  
-> ('Atomic Habits', '9780735211292', 2018);
```

```
Query OK, 3 rows affected (0.00 sec)
```

```
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

```
3 rows in set (0.00 sec)
```

Query 7: Insert 3 members

```
mysql> INSERT INTO members (full_name, email) VALUES
-> ('Anil Kumar', 'anil.kumar@example.com'),
[ -> ('Priya Sharma', 'priya.sharma@example.com'),
-> ('Ravi Verma', 'ravi.verma@example.com');
```

Query OK, 3 rows affected (0.00 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

3 rows in set (0.00 sec)

Step 4 Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

Test 2: NULL title (NOT NULL violation)

Test 3: Future publication year (CHECK violation)

Test 4: Duplicate email (UNIQUE violation)

```
mysql> INSERT INTO books (title, isbn, published_year)
[ -> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year)
[ -> VALUES (NULL, '99999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> INSERT INTO books (title, isbn, published_year)
[ -> VALUES ('Time Traveler', '88888888888888', 2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members (full_name, email)
[ -> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql>
```